"I consider mathematical quantities in this place not as consisting of very small parts, but as described by a continued motion." Isaac Newton 25 Apr 2026

The Foundation of Calculus: Limits, Local Change and Accumulation

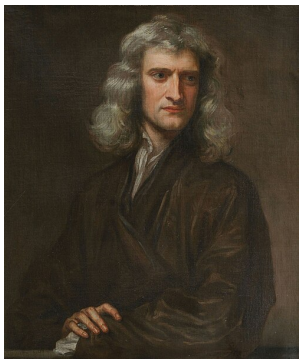
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Abstract. This exposition develops single-variable calculus through the control of approximation error. Completeness supplies limits, differentiation identifies a local linear model, and Riemann integration defines accumulation through partitions. The Mean Value Theorem connects pointwise derivative information with finite change. The Fundamental Theorem then relates the accumulation function to its integrand and yields endpoint evaluation. These results support optimisation, Newton iteration, logarithms and exponentials, integration techniques, geometric applications, and parametric and polar curves. Improper integrals and infinite series are treated as limit problems with explicit convergence conditions. Taylor approximation is separated from representation by an infinite series, with a smooth nonanalytic function showing why the remainder cannot be discarded. Reproducible calculations examine finite-difference error, quadrature convergence and root finding. An advanced appendix uses a sampled Newton basin image for a cubic polynomial to illustrate the global sensitivity of iteration, with its finite grid, stopping tolerance and iteration cap retained as part of the interpretation. The real-number completeness axiom and several compactness consequences are stated explicitly. The results concern established mathematics and numerical illustrations; the images and computations do not replace proofs.



Isaac Newton, portrait by Godfrey Kneller (1689).
Isaac Newton Institute, Cambridge; public domain.

1. Introduction and scope

A tangent line and an accumulated area appear to ask for different information. The tangent describes behaviour near one point. The area records contributions across an interval. Calculus connects them by making both constructions depend on a limit whose error can be controlled. The connection is strongest when the hypotheses are kept visible. A drawing can suggest a tangent, but a corner can prevent its slope from existing. A sequence of finer sums can suggest an area, but a bounded function need not be Riemann integrable.

The central problem here is to determine when a local approximation gives reliable information about a finite change or an accumulated total. Differentiation supplies a linear approximation with an error smaller than its input increment. The Mean Value Theorem turns this local information into a statement across an interval. Integration then asks whether the errors made in replacing a function by many simple pieces disappear together. The Fundamental Theorem explains why these two operations meet.

The same concern persists when the calculations become more elaborate. An integration substitution must account for how intervals change. An infinite series needs a limit of partial sums. A Taylor polynomial needs a remainder estimate. Newton's method can be extremely accurate near a suitable root while behaving poorly elsewhere. None of these distinctions is visible from a formula alone.

The prerequisites are algebra, elementary functions, plane geometry, trigonometry and the arithmetic of inequalities. Limits and one-variable derivatives and integrals are developed within the exposition. The ordered-field properties and completeness of $\mathbb{R}$ are assumed. Several consequences of compactness are stated with their roles identified rather than replaced by a construction of the real numbers. The standard real-variable development ends with the conclusions. Appendix A is an optional advanced extension to Newton dynamics in the complex plane. It introduces the complex arithmetic needed to iterate a polynomial and does not form part of the prerequisites for the main text.

The source courses supply complementary viewpoints. The main single-variable course provides the computational range. The revisited course sharpens the interpretation of limiting operations. The visual lessons motivate local change and accumulation, while the real-analysis lectures provide precise hypotheses and counterexamples. Source attribution and a guide to the downloaded transcript archive appear in the bibliography. The development here is a connected exposition rather than a sequence of lecture summaries.

1.1. Notation and the status of an approximation

The independent variable is usually x. An increment h is a nonzero real number that may have either sign. The notation $h \to 0$ describes a limiting process; it does not give h the value zero inside a quotient. A limit is evaluated within the domain of the function. Unless specified otherwise, an interval is a real interval and angles are measured in radians.

We write $r(h) = o(h)$ when $r(h)/h \to 0$. We write $r(h) = O(h^p)$ near zero if $|r(h)| \le C|h|^p$ there for some constant C independent of h. The first notation expresses a vanishing relative error; the second gives a bound on its size. Neither notation by itself identifies the numerical value of the error.

$$r(h) = o(h) \iff \lim_{h\to 0} \frac{r(h)}{h} = 0, \qquad |r(h)| \le C|h|^p \implies r(h) = O(h^p). \tag{1}$$

An exact equality, an asymptotic statement and a computed decimal have different meanings. These distinctions will be retained even when their graphs look indistinguishable at the displayed scale.

2. Real numbers and completeness

2.1. An ordered field does not yet supply every limit

Rational arithmetic permits addition, multiplication and division by nonzero numbers. It also permits order comparisons. Yet a rational sequence can approach a number absent from the rationals. To see the obstruction without relying on a decimal picture, suppose $\sqrt{2} = p/q$ for coprime integers $p, q > 0$. Then $p^2 = 2q^2$, so p is even. Writing $p = 2k$ gives $q^2 = 2k^2$, so q is also even. This contradicts coprimality.

The calculation does not say that the equation $x^2 = 2$ has no real solution. It says that rational arithmetic alone has a gap relevant to a limiting construction. Calculus needs a number system in which suitable approximations can have limits within the system.

An upper bound of a set S is a number b with $s \le b$ for every $s \in S$. Its least upper bound, written $\sup S$, is the smallest such number. Completeness asserts that every nonempty subset of $\mathbb{R}$ bounded above has a real least upper bound. This is an axiom of the real-number model used here. It is not being proved from a picture of the number line.

2.2. Monotone approximation

Theorem 2.1 (Bounded monotone convergence)**.** *Every increasing real sequence bounded above converges to the supremum of its terms. Every decreasing real sequence bounded below also converges.*

Proof. Let (a_n) be increasing and bounded above, and let $L = \sup\{a_n : n \in \mathbb{N}\}$. For any $\epsilon > 0$, the number $L - \epsilon$ cannot be an upper bound. Some a_N therefore exceeds it. Monotonicity gives, for every $n \ge N$,

$$L - \epsilon < a_N \le a_n \le L, \qquad |a_n - L| < \epsilon. \tag{2}$$

Thus an increasing bounded sequence converges. A decreasing bounded sequence follows by changing signs. The proof identifies where completeness enters: it supplies L before the approximation argument begins. □

For $0 < q < 1$, the geometric partial sums are increasing and bounded. Their finite algebraic identity also gives the error explicitly:

$$\sum_{k=0}^{n} q^k = \frac{1-q^{n+1}}{1-q}, \qquad \frac{1}{1-q} - \sum_{k=0}^{n} q^k = \frac{q^{n+1}}{1-q}. \tag{3}$$

The right-hand error tends to zero. This conclusion concerns a sequence of finite sums. It does not require performing infinitely many additions as a completed arithmetic procedure.

2.3. Cauchy sequences and interval consequences

A sequence is Cauchy if its sufficiently late terms are as close to one another as any prescribed tolerance requires:

$$\forall \epsilon > 0 \ \exists N \ \forall m, n \geq N : \quad |a_n - a_m| < \epsilon. \tag{4}$$

Every convergent sequence is Cauchy by the triangle inequality. The converse holds in $\mathbb{R}$ and is an equivalent form of completeness. In $\mathbb{Q}$ the converse can fail, because the approximations may approach a missing irrational value. We use the real Cauchy convergence theorem as an established consequence of the least-upper-bound axiom.

Three further consequences will be needed. A continuous function on a closed bounded interval attains a maximum and minimum, takes every intermediate value, and is uniformly continuous there. Their proofs can be organised using convergent subsequences or nested intervals. The derivative and integration arguments below will state exactly where these interval theorems are used.

The closed and bounded qualification matters. The function $1/x$ is continuous on $(0, 1)$ but is unbounded, while x on $(0, 1)$ has neither a largest nor a smallest value. Continuity at each individual point cannot supply missing endpoints or control an unbounded domain.

3. Limits and continuity

3.1. The order of the quantifiers

To say that $f(x)$ tends to L as x tends to a is to require that every output tolerance can be met by restricting the input sufficiently. The point a must be an accumulation point of the domain, so there are domain points arbitrarily near it other than itself. The definition is

$$\lim_{x \to a} f(x) = L \quad \Longleftrightarrow \quad \forall \epsilon > 0 \ \exists \delta > 0 : \ 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon. \tag{5}$$

The chosen δ may depend on ϵ and a, but it must work for every admissible x in that neighbourhood. Choosing a new δ after seeing each individual x would not establish the stated limit. The puncture $0 < |x - a|$ separates nearby behaviour from the value assigned at a.

Example 3.1 (An explicit tolerance bound)**.** For $f(x) = x^2$ at $a = 2$, factor the error. First impose $|x - 2| < 1$, so $|x + 2| < 5$. Then

$$|x^2 - 4| = |x - 2||x + 2| < 5|x - 2|, \qquad \delta = \min\{1, \epsilon/5\}. \tag{6}$$

This single choice proves the limit for every positive ϵ. It is a sufficient neighbourhood, not necessarily the largest possible one. A proof needs a working bound, not the optimal bound.

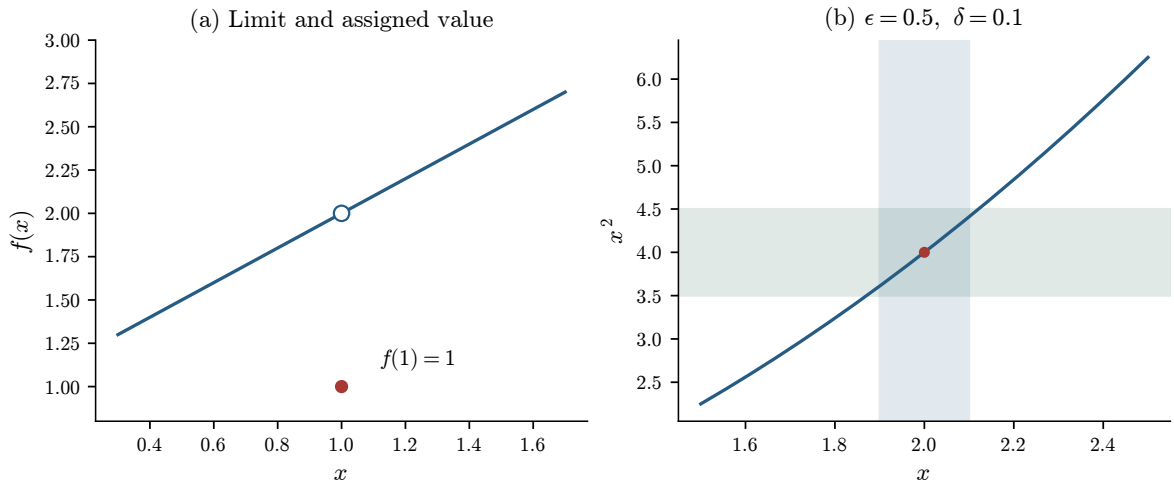


Figure 1. Calculated limit examples. Left: $(x^2 - 1)/(x - 1)$ for $x \neq 1$, with the value at 1 deliberately assigned to be 1. The limit is still 2. Right: an input neighbourhood guaranteeing the displayed output tolerance for x^2 at 2. The graphs illustrate the definitions; the bounds in the text establish them.

3.2. Uniqueness and the rules for combining limits

Theorem 3.2 (Uniqueness of a finite limit)**.** *A function has at most one finite limit at any accumulation point of its domain.*

Proof. If $L \neq M$, choose tolerance $|L - M|/3$ around each proposed limit and restrict to a common neighbourhood. The triangle inequality would give

$$|L - M| \leq |L - f(x)| + |f(x) - M| < \frac{2}{3}|L - M|, \tag{7}$$

which is impossible. Nearby domain points exist by the accumulation-point assumption. □

If $f \to L$ and $g \to M$, the sum tends to $L + M$ because the two errors can each be bounded by half the desired tolerance. For the product, write

$$f(x)g(x) - LM = f(x)[g(x) - M] + M[f(x) - L]. \tag{8}$$

A function with a finite limit is bounded in some punctured neighbourhood. This bounds the factor $f(x)$, after which the two errors can again be controlled. Quotients are handled by keeping the denominator separated from zero; the condition is $M \neq 0$. Substitution into a continuous outer function follows from applying its input tolerance to the inner function's output error.

The squeeze theorem uses order instead. If $u(x) \leq f(x) \leq v(x)$ near a, and both outer functions tend to L, their tolerance bounds trap f in the same band. For example, $|x \sin(1/x)| \leq |x|$ proves a zero limit despite continued oscillation of the sine factor.

3.3. A missing value, a jump and an unbounded limit

For $x \neq 1$, $(x^2-1)/(x-1) = x+1$. Its limit at 1 is 2, regardless of whether the original quotient is undefined there or a separate value is assigned. Defining the value to be 2 repairs the discontinuity. A jump cannot be repaired this way because its one-sided limits disagree.

One-sided limits restrict the approach to $x > a$ or $x < a$. A two-sided finite limit exists precisely when both one-sided limits exist and agree, for a domain containing a punctured interval. The function $|x|/x$ has right limit 1 and left limit -1 at zero. No assignment at zero reconciles them.

The notation $f(x) \to +\infty$ is a statement of unbounded growth, not convergence to a real number. For every $M > 0$, a sufficiently small punctured neighbourhood must give $f(x) > M$. Thus $1/x^2 \to +\infty$ at zero, while $1/x$ has opposite infinite one-sided limits. At infinity, the input condition is instead $x > A$ for sufficiently large A.

3.4. Continuity and uniform continuity

A function is continuous at a when it is defined there and its limit there equals $f(a)$. In the tolerance definition the puncture can then be removed. Polynomials are continuous by the limit rules, and rational functions are continuous wherever their denominators are nonzero.

Continuity is local. Uniform continuity on a set strengthens it by choosing one input tolerance that works at every point:

$$\forall \epsilon > 0 \; \exists \delta > 0 \; \forall x, y : \quad |x - y| < \delta \implies |f(x) - f(y)| < \epsilon. \tag{9}$$

For x^2 on $[0, 1]$, the bound $|x^2 - y^2| \leq 2|x - y|$ gives a uniform choice. On all of $\mathbb{R}$, no fixed δ works for a fixed small output tolerance, because $|x^2 - y^2| = |x - y||x + y|$ grows with the location. The compact-interval theorem stated earlier guarantees uniform continuity for every continuous function on $[a, b]$. That common control will allow the errors from many integration subintervals to be bounded at once.

4. The derivative as a local linear model

4.1. Difference quotients and the remainder

Fix an interior point a of the domain. The difference quotient is the slope between two distinct points of the graph. If its limit is finite, that limit defines the derivative:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}. \tag{10}$$

The quotient is evaluated for $h \neq 0$. Its limit does not arise by substituting zero into a zero denominator. Multiplying the defining error by h gives an equivalent statement:

$$f(a+h) = f(a) + f'(a)h + r(h), \qquad \frac{r(h)}{h} \to 0. \tag{11}$$

This is the local linear model. The graph differs from its tangent line by an amount small compared with $|h|$. Requiring only $r(h) \to 0$ would give continuity, not differentiability. A corner such as $|x|$ has a small change in value but no single linear coefficient that works on both sides.

Theorem 4.1 (Uniqueness of the linear model and continuity)**.** *If f is differentiable at an interior point a, its local linear coefficient is unique and f is continuous at a.*

Proof. If both A and B gave remainders of order $o(h)$, subtracting the two representations and dividing by h would give $A - B \to 0$. Since $A - B$ is constant, it must be zero. Differentiability also implies continuity because both $f'(a)h$ and $r(h)$ tend to zero. □

4.2. Two elementary calculations

Example 4.2 (Square and reciprocal)**.** For the square function, expansion is exact:

$$(a+h)^2 - a^2 = 2ah + h^2, \qquad \frac{(a+h)^2 - a^2}{h} = 2a + h \to 2a. \tag{12}$$

Its linearisation is $a^2 + 2ah$, and its remainder is h^2. For $f(x) = 1/x$ with $a \neq 0$,

$$\frac{1/(a+h) - 1/a}{h} = -\frac{1}{a(a+h)} \to -\frac{1}{a^2}. \tag{13}$$

The denominator stays nonzero for sufficiently small h. Both computations use an algebraic simplification valid away from $h = 0$, followed by a limit. Neither discards a nonzero term without explaining why its contribution tends to zero.

If x has dimensions of time and f of distance, the derivative has dimensions of speed. A second derivative has dimensions of acceleration. In a purely mathematical graph, the same construction measures sensitivity without requiring a physical motion.

4.3. A derivative need not be continuous

The left and right difference quotients of $|x|$ at zero are -1 and 1, so the derivative fails there. A different obstruction appears for

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases} \tag{14}$$

At zero, the quotient is $h \sin(1/h)$, which tends to zero by squeezing. Thus $f'(0) = 0$. At nonzero x, the rules established in the next section give

$$f'(x) = 2x \sin(1/x) - \cos(1/x). \tag{15}$$

This derivative does not tend to zero at zero. The original function is differentiable everywhere, but its derivative is discontinuous. Requiring a continuous derivative is an additional hypothesis, often useful for estimates, rather than part of the definition of a derivative.

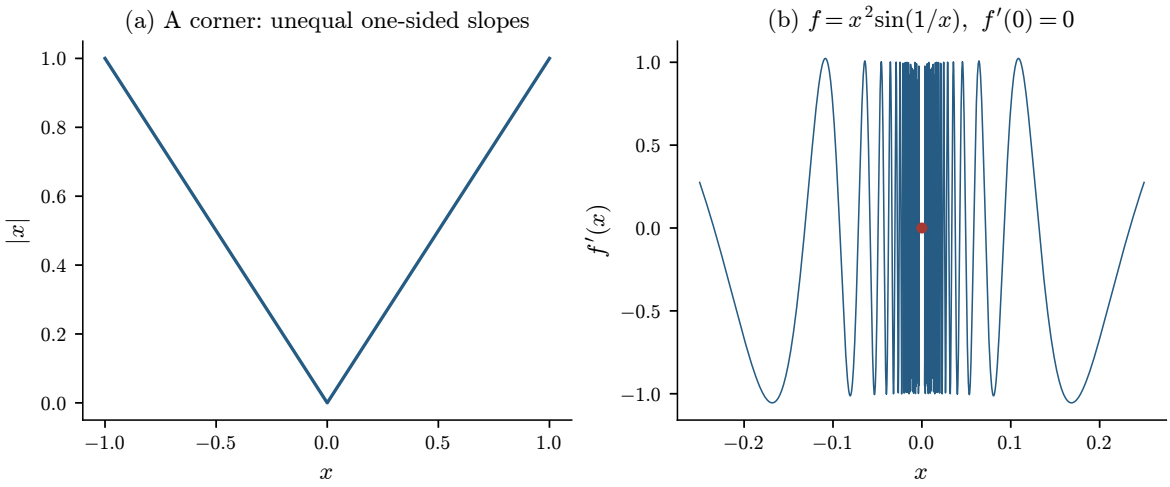


Figure 2. Calculated regularity examples. Left: $|x|$ is continuous but has unequal one-sided slopes at zero. Right: the derivative of the oscillatory function in the text, with $f'(0) = 0$ marked. Finite sampling cannot display every oscillation near zero; the formula and the sequence argument determine its limiting behaviour.

For instance, at $x_n = 1/(2\pi n)$ the derivative equals -1, while at $y_n = 1/((2n+1)\pi)$ it equals 1. Both sequences approach zero. The two incompatible limiting values prove the derivative's discontinuity independently of the displayed graph.

5. Rules for differentiating functions

5.1. Sums, products and quotients

The linear model provides the differentiation rules without treating the derivative as an isolated list of formulas. If u and v are differentiable at a, write their increments as

$u'(a)h + o(h)$ and $v'(a)h + o(h)$. Adding the models or multiplying them by a constant gives the sum and scaling rules.

For a product, an exact identity separates the first-order increments from their interaction:

$$\Delta(uv) = u\,\Delta v + v\,\Delta u + \Delta u\,\Delta v. \tag{16}$$

Both increments are $O(h)$, so their product is $O(h^2)$ and becomes zero after division by h and passage to the limit. Therefore

$$(u+v)' = u' + v', \qquad (cu)' = cu', \qquad (uv)' = u'v + uv'. \tag{17}$$

For a reciprocal, the same difference-quotient calculation used for $1/x$ gives $(1/v)' = -v'/v^2$ where $v \neq 0$. Combining it with the product rule yields

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \qquad v \neq 0. \tag{18}$$

The denominator condition belongs to the rule. A cancellation in a simplified expression may permit an extension to a missing point, but that is a separate function and must be defined there.

5.2. Composition and the chain rule

Theorem 5.1 (Chain rule)**.** *If g is differentiable at a and f is differentiable at $g(a)$, their composition is differentiable at a with derivative $f'(g(a))g'(a)$.*

Proof. Put $k = g(a+h) - g(a)$. Differentiability of g gives $k = g'(a)h + o(h)$, so $k = O(h)$. The outer function has the model

$$f(g(a)+k) = f(g(a)) + f'(g(a))k + k\rho(k), \qquad \rho(k) \to 0. \tag{19}$$

Define $\rho(0) = 0$ if necessary. This keeps the formula valid even for increments at which $g(a+h) = g(a)$. Substituting the model for k, the final term divided by h is $(k/h)\rho(k) \to 0$. It follows that

$$(f \circ g)'(a) = f'(g(a))g'(a). \tag{20}$$

□

This argument avoids dividing by the intermediate change k, which may vanish for nonzero h. The inner derivative measures how the intermediate variable changes; the outer derivative measures sensitivity to that intermediate variable. Their product measures the complete first-order response.

Example 5.2 (A composed power)**.** For example, $(1+x^2)^5$ has derivative $10x(1+x^2)^4$. The power rule applies to the outer function while $2x$ accounts for the inner change. Treating the entire expression as a fifth power of x would omit that dependence.

5.3. Powers and circular functions

For a positive integer n, the binomial expansion gives $(x+h)^n - x^n = nx^{n-1}h + O(h^2)$. Division by h proves the power rule. Negative integer powers follow from reciprocals, and rational powers follow on intervals where the corresponding real branch is defined. A later construction using logarithms extends the rule to arbitrary real powers of positive x.

$$\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}. \tag{21}$$

The sine derivative depends on the radian normalization. Elementary unit-circle geometry gives, for $0 < h < \pi/2$, $\sin h < h < \tan h$. Dividing by $\sin h$ and squeezing gives $\sin h/h \to 1$. Symmetry supplies the negative approach. Also,

$$\frac{1-\cos h}{h} = \frac{\sin^2 h}{h(1+\cos h)} \to 0. \tag{22}$$

The angle-addition identities then reduce the sine and cosine difference quotients to these two limits:

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x, \qquad (\tan x)' = \frac{1}{\cos^2 x}. \tag{23}$$

The last formula applies where $\cos x \neq 0$. If a numerical angle were expressed in degrees, differentiating with respect to that degree coordinate would introduce a factor $\pi/180$. Radians make the geometric and analytic normalization agree.

5.4. Inverse functions and implicit relations

A continuous strictly monotone function on an interval has a continuous inverse on its image. Suppose $f'(a) \neq 0$. Nearby inverse difference quotients are reciprocals of the corresponding nonzero quotients of f, so

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}. \tag{24}$$

The continuity of the inverse ensures that approaching $f(a)$ corresponds to approaching a. The formula is not a proof that an arbitrary function has an inverse. Restricting sine to $[-\pi/2, \pi/2]$ gives the inverse branch arcsin. At its interior points,

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1; \qquad (\arctan x)' = \frac{1}{1+x^2}. \tag{25}$$

An equation can instead describe a branch implicitly. On a differentiable branch of $x^2 + y^2 = R^2$, regard y as a function of x and differentiate both sides:

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y} \quad (y \neq 0). \tag{26}$$

At the circle's leftmost and rightmost points, $y = 0$ and the tangent is vertical. The failure of a finite derivative with respect to x there does not mean that the geometric curve has a corner. It means that this local graph coordinate is unsuitable for a finite slope.

6. From local slopes to finite change

6.1. An interior extremum

Theorem 6.1 (Fermat's stationary-point condition)**.** *If a differentiable function has a local extremum at an interior point c, then $f'(c) = 0$.*

Proof. Suppose f has a local maximum at c. For small positive h, the quotient $[f(c+h) - f(c)]/h$ is nonpositive. For negative h, it is nonnegative. A common finite limit must therefore be zero. The minimum case follows by changing signs. □

Remark 6.2 (Stationarity is a necessary condition)*.* A zero derivative is necessary at a differentiable interior extremum, not sufficient. The function x^3 has derivative zero at zero and is increasing through it. Endpoints and nondifferentiable points must also be examined in an optimisation problem.

6.2. Rolle's theorem and the Mean Value Theorem

Theorem 6.3 (Rolle's theorem)**.** *Let $a < b$. If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, there is a point $c \in (a, b)$ with $f'(c) = 0$.*

Proof. The extreme-value theorem supplies a maximum and minimum. If the function is not constant, at least one extremum occurs in the interior. Fermat's condition gives $f'(c) = 0$ there. If the function is constant, any interior point works. □

Theorem 6.4 (Mean Value Theorem)**.** *Let $a < b$. If f is continuous on $[a, b]$ and differentiable on (a, b), there is a point $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}. \tag{27}$$

Proof. Subtract the line joining $(a, f(a))$ and $(b, f(b))$. The resulting function is continuous on $[a, b]$, differentiable on (a, b), and has equal endpoint values. Rolle's theorem gives a point where its derivative is zero, which is the asserted identity. □

Remark 6.5 (The hypotheses of the Mean Value Theorem)*.* Continuity on the closed interval and differentiability on its interior are both required. Applying the conclusion to $|x|$ on $[-1, 1]$ would ask for derivative zero somewhere, although every existing derivative there is -1 or 1. The missing derivative at zero is exactly the failed hypothesis.

6.3. Monotonicity and a bound on error

If $f' > 0$ throughout an interval, applying the theorem between any two distinct points proves that f is strictly increasing. If $f' \geq 0$, the conclusion is nondecreasing. If $f' = 0$ everywhere on an interval, any two function values agree, so f is constant there. This last conclusion explains why antiderivatives of one function differ only by a constant on a connected interval.

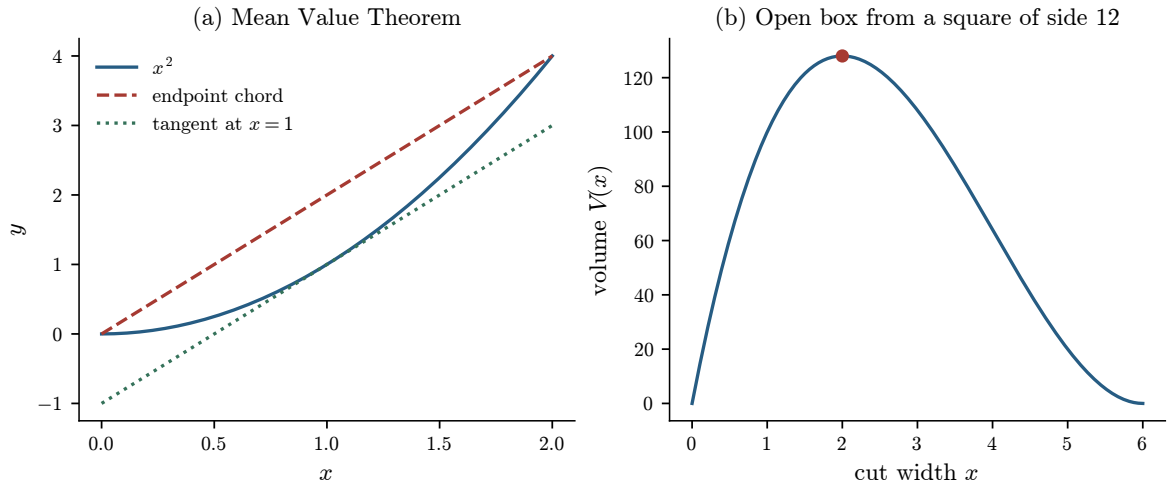


Figure 3. Calculated applications of derivative information. Left: the chord of x^2 on $[0, 2]$ is parallel to the tangent at 1, as the Mean Value Theorem predicts. Right: the volume function for the open-box construction in the text. The endpoint comparison establishes the displayed maximum.

The theorem also turns a derivative bound into a finite-change bound. If $|f'| \le K$ between two points and the endpoint continuity hypotheses hold, then

$$|f(b) - f(a)| \le K|b - a|. \tag{28}$$

This is a Lipschitz estimate. Unlike a statement about one tangent, it controls an entire interval. A large derivative can amplify input error; a small derivative bound limits the amplification. The estimate supplies a mathematical reason to inspect the domain over which an approximation will be used.

6.4. Concavity and the shape of a graph

If $f'' > 0$ on an interval, the first derivative is increasing there. The graph is concave upward: its slopes increase as the input increases. The Mean Value Theorem then places the graph above each tangent within that interval. For $f'' < 0$, the slopes decrease and the graph lies below its tangents. An inflection point requires a change of concavity. A zero second derivative alone is insufficient; x^4 has $f''(0) = 0$ while remaining concave upward on both sides.

A curve sketch can therefore combine distinct pieces of information: the domain and its excluded points, intercepts, limits near boundaries and at infinity, signs of f' and f'', and values at critical points. A finite plot may hide a narrow feature or a vertical asymptote. For $1/x$, neither an apparent connecting line across zero nor an arbitrary plotting cutoff belongs to the function. The limiting and derivative information determines which features a faithful sketch must retain.

6.5. Optimisation and related rates

Example 6.6 (The maximum-volume open box)**.** Cut squares of side x from the corners of a square sheet of side 12 and fold the remaining edges upward. In consistent length units the volume is $V = x(12 - 2x)^2$, with physical range $0 \le x \le 6$. Its derivative is

$$V'(x) = (12 - 2x)(12 - 6x). \tag{29}$$

The only interior stationary point is $x = 2$. The derivative is positive before it and negative afterwards. The endpoint volumes vanish, so $V(2) = 128$ is the global maximum on the stated interval. The domain restriction prevents an algebraic continuation with negative side lengths from entering the physical comparison.

Example 6.7 (An expanding sphere)**.** For related rates, the relation between quantities is differentiated before numerical values are inserted. A sphere with radius $r(t)$ has volume $V(t) = 4\pi r(t)^3/3$, hence

$$\frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}. \tag{30}$$

At $r = 2$ length units and $\mathrm{d}r/\mathrm{d}t = 3$ length units per time unit, the rate is 48π volume units per time unit. Replacing r by its instantaneous value before differentiating would turn a changing quantity into a constant and lose the rate being sought.

7. Linear approximation and Newton iteration

7.1. The size of a neglected term

The approximation $f(a + h) \simeq f(a) + f'(a)h$ becomes informative only when its remainder is controlled over the required range. For $f(x) = \sqrt{x}$ at $a = 4$, it gives $\sqrt{4 + h} \simeq 2 + h/4$. The derivative exists only for positive arguments, and the approximation is local. Taylor's theorem later gives a quantitative second-order remainder.

At an extremum a first-order model may be constant, so second-order information becomes decisive. If f has a continuous second derivative near c and $f'(c) = 0$, then

$$f(c + h) - f(c) = \frac{1}{2} f''(c)h^2 + o(h^2). \tag{31}$$

Thus $f''(c) > 0$ gives a strict local minimum, while $f''(c) < 0$ gives a strict local maximum. When the second derivative is zero, the test is inconclusive. The functions x^4, $-x^4$ and x^3 show all three possible behaviours at zero.

7.2. Turning the tangent into a root estimate

Suppose a current estimate x_n is near a zero of f. Replace the graph by its tangent at x_n and find where that line meets the horizontal axis. Solving $0 = f(x_n) + f'(x_n)(x_{n+1} - x_n)$ gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \qquad f'(x_n) \neq 0. \tag{32}$$

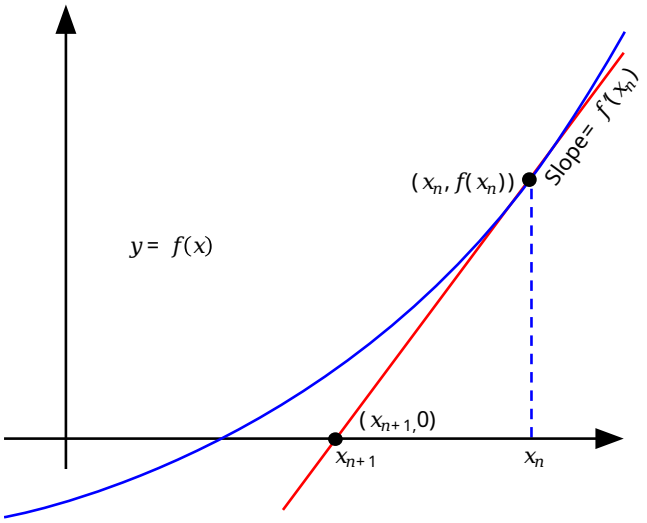


Figure 4. Newton's tangent construction. The current point $(x_n, f(x_n))$ determines a tangent whose horizontal intercept is x_{n+1}. Vector illustration by Oleg Alexandrov, redrawn by Pbroks13 with later labels by Shubjt, via Wikimedia Commons; public domain. The drawing is schematic, not a computed trajectory for a specified function.

Example 7.1 (Newton iteration for the square root of two)**.** For $f(x) = x^2 - 2$, the update becomes $(x_n + 2/x_n)/2$. From $x_0 = 1$, its first values are 1.5, 1.416666... and 1.414215686.... For any positive starting value the first update is at least $\sqrt{2}$, because

$$\frac{1}{2}\left(x + \frac{2}{x}\right) - \sqrt{2} = \frac{(x - \sqrt{2})^2}{2x} \geq 0. \tag{33}$$

Above $\sqrt{2}$ the update is smaller than x while remaining above the root. Completeness therefore supplies a limit, and the fixed-point equation identifies it as $\sqrt{2}$. The exact error equation is

$$e_{n+1} = \frac{e_n^2}{2x_n}, \qquad e_n = x_n - \sqrt{2}. \tag{34}$$

Once the error is small, squaring it explains the rapid gain in accuracy. This is an exact relation in real arithmetic. Floating-point rounding eventually limits the computed improvement.

7.3. Local convergence and global failure

For a general simple root α, assume f is twice continuously differentiable nearby and $f'(\alpha) \neq 0$. Taylor's theorem applied between x_n and α gives

$$e_{n+1} = \frac{f''(\xi_n)}{2f'(x_n)} e_n^2, \tag{35}$$

where ξ_n lies between them. In a sufficiently small neighbourhood, $|f'| \geq m > 0$ and $|f''| \leq M$. Then $|e_{n+1}| \leq M|e_n|^2/(2m)$. Choosing the initial error small enough keeps the iterates in the neighbourhood and makes the errors tend to zero at least quadratically. If the leading coefficient vanishes, convergence can be faster.

These local hypotheses do not guarantee convergence from every starting point. A horizontal tangent makes the update undefined. For $f(x) = x^3 - 2x + 2$, starting at zero gives the exact cycle $0, 1, 0, 1, \ldots$. For a root of multiplicity $m > 1$, the model $f(x) = (x-\alpha)^m$ instead gives $e_{n+1} = (1-1/m)e_n$, which is only linear convergence. The optional basin picture in Appendix A explores how global dependence on the starting point can remain complicated even when each root is locally attracting.

8. The definite integral as a controlled sum

8.1. Partitions, tags and mesh

Let f be bounded on $[a, b]$, where $a < b$. A partition consists of points $a = x_0 < x_1 < \cdots < x_n = b$. Choose a tag $\xi_i \in [x_{i-1}, x_i]$ in each subinterval. Its Riemann sum is

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i, \qquad \Delta x_i = x_i - x_{i-1}, \qquad \|P\| = \max_i \Delta x_i. \tag{36}$$

The mesh $\|P\|$ measures the largest remaining interval. Merely increasing the number of intervals is insufficient if one interval remains large. The Riemann integral is I if every sufficiently fine tagged partition has its sum close to I:

$$\forall \epsilon > 0 \; \exists \delta > 0 : \quad \|P\| < \delta \implies |S(f, P, \xi) - I| < \epsilon \quad \text{for all tags.} \tag{37}$$

The requirement covers arbitrary partitions, not only one selected sequence of equal subdivisions or one convenient set of sample points. This is how the definition prevents the sampling choice from deciding the answer.

8.2. Lower and upper sums

Let m_i and M_i be the infimum and supremum of f on the ith subinterval. Then every tagged sum lies between

$$L(f, P) = \sum_i m_i \Delta x_i, \qquad U(f, P) = \sum_i M_i \Delta x_i. \tag{38}$$

Refining a partition raises its lower sum and lowers its upper sum. Completeness supplies the supremum of all lower sums and the infimum of all upper sums. They agree precisely when the bounded function is Riemann integrable. Equivalently, for every $\epsilon > 0$ there is a partition for which $U(f, P) - L(f, P) < \epsilon$. This is the Darboux criterion, used here in its equivalent form of the Riemann definition.

Theorem 8.1 (Continuous functions are Riemann integrable)**.** *Every continuous real-valued function on a closed bounded interval $[a, b]$, with $a < b$, is Riemann integrable.*

Proof. The extreme-value theorem gives boundedness. Uniform continuity ensures that every sufficiently short subinterval has oscillation smaller than $\epsilon/(b-a)$. Consequently,

$$U(f,P) - L(f,P) = \sum_i (M_i - m_i)\Delta x_i < \frac{\epsilon}{b-a} \sum_i \Delta x_i = \epsilon. \tag{39}$$

Thus every continuous function on a closed bounded interval is Riemann integrable. The crucial step is that one interval length works everywhere, allowing all the small errors to be controlled together. □

8.3. A limit that can be computed directly

Example 8.2 (Integrating a square directly from sums)**.** For x^2 on $[0,1]$, equal subintervals give a lower sum using left endpoints and an upper sum using right endpoints. The finite sum-of-squares identity gives

$$L_n = \frac{(n-1)n(2n-1)}{6n^3}, \qquad U_n = \frac{n(n+1)(2n+1)}{6n^3}, \qquad U_n - L_n = \frac{1}{n}. \tag{40}$$

Both sums tend to $1/3$, so their common value is the integral. The exact gap $1/n$ quantifies the bracketing error. The answer has not been obtained by presuming an antiderivative formula before integration is defined.

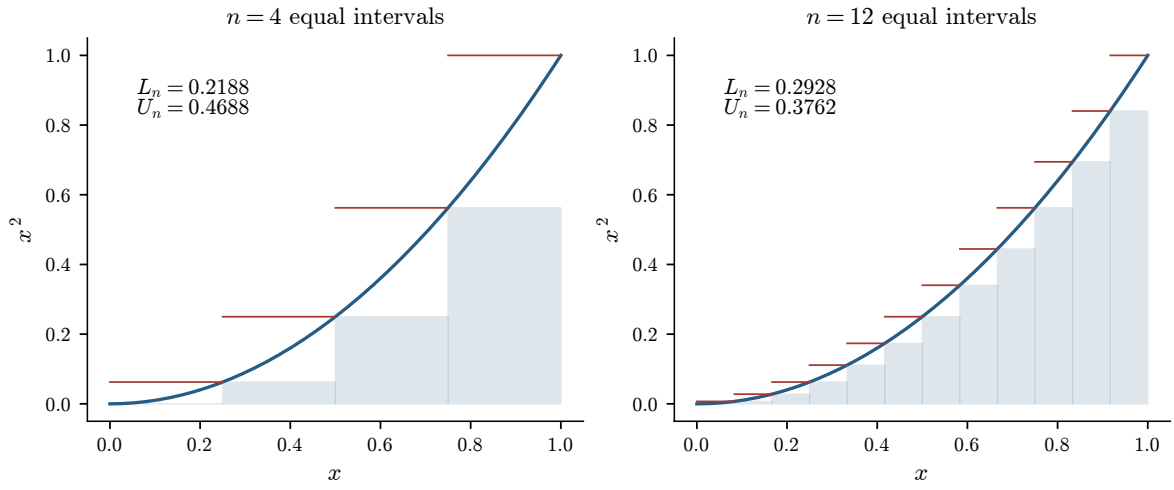


Figure 5. Calculated lower and upper sums for x^2 on $[0,1]$. Shaded rectangles use left endpoints, and red top edges mark the upper rectangles. Increasing the subdivision count narrows the exact upper–lower gap from $1/4$ to $1/12$.

8.4. Integrability is a real condition

Example 8.3 (A bounded function that is not Riemann integrable)**.** Define $D(x) = 1$ for rational x and $D(x) = 0$ for irrational x on $[0,1]$. Every nontrivial interval contains both kinds of number. Every lower sum is zero and every upper sum is one. The function

is bounded but fails the Darboux criterion. Sampling only rational or only irrational tags would produce incompatible answers.

Remark 8.4 (Integrable discontinuities). By contrast, changing a bounded integrable function at finitely many points does not change its integral. Those points can be covered by intervals of arbitrarily small total length, and the additional upper–lower contribution is bounded by that length times a fixed height bound. A finite jump can therefore be compatible with integrability. Continuity is sufficient, but it is not necessary.

For nonnegative f, the integral agrees with the area under the graph. In general it is signed accumulation. Reversing endpoints changes the sign, and additivity and linearity follow by combining the corresponding sums:

$$\int_a^b f\,\mathrm{d}x = -\int_b^a f\,\mathrm{d}x, \qquad \int_a^c f\,\mathrm{d}x + \int_c^b f\,\mathrm{d}x = \int_a^b f\,\mathrm{d}x. \tag{41}$$

The symbol x in a definite integral is a bound variable. Replacing it consistently by t changes no value. An indefinite integral, by contrast, denotes a family of antiderivatives. Their relationship is the subject of the next section.

9. The Fundamental Theorem of Calculus

9.1. Differentiating an accumulation function

Theorem 9.1 (Fundamental Theorem: differentiation of accumulation)**.** *Let f be continuous on $[a,b]$, with $a<b$, and put $F(x)=\int_a^x f(t)\,\mathrm{d}t$. Then F is differentiable at every interior point and $F'(x)=f(x)$.*

Proof. For an interior point x and a small nonzero h, additivity gives

$$\frac{F(x+h)-F(x)}{h} - f(x) = \frac{1}{h}\int_x^{x+h} [f(t)-f(x)]\,\mathrm{d}t. \tag{42}$$

For either sign of h, the absolute value is at most the largest value of $|f(t)-f(x)|$ between the two endpoints. Continuity makes that bound tend to zero. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x). \tag{43}$$

□

The accumulation function has a derivative even if no elementary expression for the integral is known. For instance, $\int_0^x e^{-t^2}\,\mathrm{d}t$ will define a valid differentiable function once the exponential is constructed. An elementary antiderivative is a matter of available notation, not a prerequisite for existence.

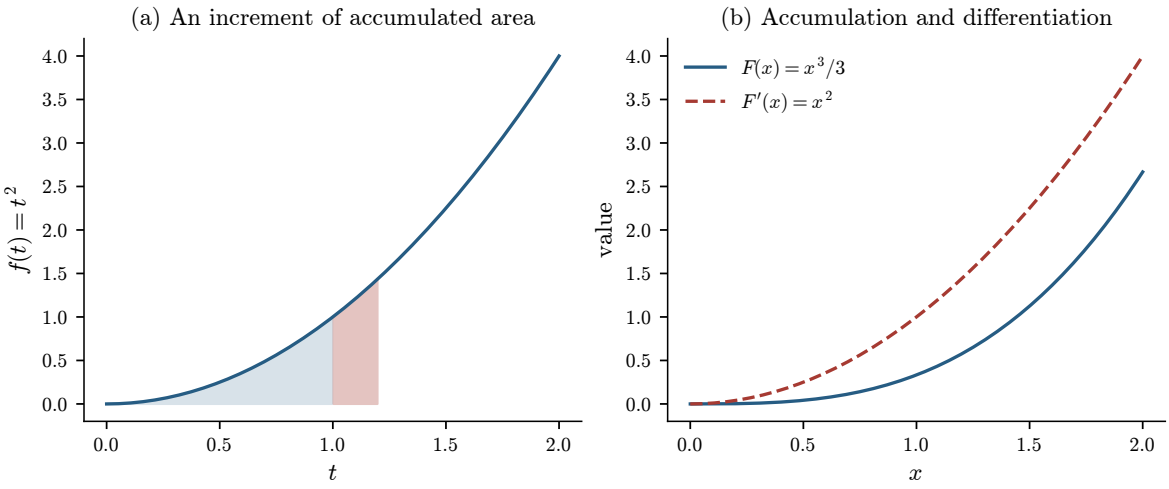


Figure 6. Calculated accumulation example for $f(t) = t^2$. Left: increasing the upper endpoint from 1 to 1.2 adds the shaded strip. Its displayed width is finite; the derivative argument takes a limit as the width tends to zero. Right: $F(x) = x^3/3$ and its derivative.

9.2. Evaluating an integral from any antiderivative

Theorem 9.2 (Fundamental Theorem: endpoint evaluation)**.** *Let f be continuous on $[a, b]$, with $a < b$. If G is continuous on $[a, b]$, differentiable on (a, b), and $G' = f$ there, then*

$$\int_a^b f(x)\,\mathrm{d}x = G(b) - G(a). \tag{44}$$

Proof. The accumulation function F is continuous on $[a, b]$: boundedness of f gives $|F(y) - F(x)| \le K|y - x|$ for a bound $|f| \le K$. Theorem 9.1 gives $F' = f$ on the interior. The Mean Value Theorem then makes $G - F$ constant on $[a, b]$. Since $F(a) = 0$, comparing its endpoint values proves the formula. □

This endpoint formula is the evaluation form of the Fundamental Theorem. The proof joins three ingredients that were previously distinct: existence of the integral, differentiation of accumulation, and constancy of a function whose derivative is zero. The numbering of the theorem's two forms varies among textbooks; their content is what matters here.

For x^2, the antiderivative $x^3/3$ reproduces the value $1/3$ previously obtained from sums. In general, finding an antiderivative makes evaluation efficient, but it does not replace checking that the integral's interval and singularities satisfy the hypotheses.

9.3. Variable endpoints and a net change

If u and v are differentiable and f is continuous on the required range, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t = f(v(x))v'(x) - f(u(x))u'(x). \tag{45}$$

Example 9.3 (Two moving endpoints)**.** For example, differentiating $\int_{x^2}^{\sin x}(1+t^2)\,\mathrm{d}t$ gives $(1+\sin^2 x)\cos x - 2x(1+x^4)$. Both moving endpoints contribute.

The integrand here depends on its integration variable alone; differentiating an additional parameter inside the integrand would require a further theorem.

Remark 9.4 (Net change and total accumulation)*.* The same endpoint relation turns a rate into a net change. If $s' = v$ and the hypotheses hold, then $s(b) - s(a) = \int_a^b v(t)\,\mathrm{d}t$. Total distance is instead $\int_a^b |v(t)|\,\mathrm{d}t$. A reversal of direction can cancel displacement while still contributing positive distance.

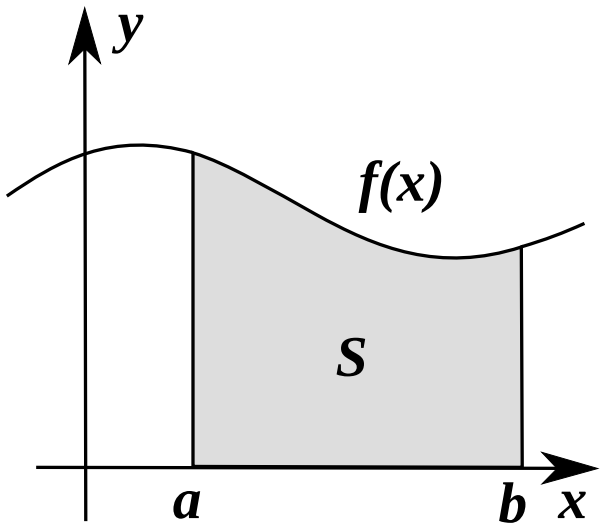


Figure 7. Geometric interpretation of a definite integral for a nonnegative function. The shaded region has area $S = \int_a^b f(x)\,\mathrm{d}x$. Vector artwork by 4C, with revisions by Oleg Alexandrov and Antonsusi, via Wikimedia Commons, CC BY-SA 3.0. No specific formula for the schematic curve is asserted.

10. Logarithms, exponentials and differential equations

10.1. A function defined by its accumulation

The Fundamental Theorem permits a construction of the logarithm that does not assume its derivative in advance. For $x > 0$, define

$$\ln x = \int_1^x \frac{1}{t}\,\mathrm{d}t, \qquad (\ln x)' = \frac{1}{x}. \tag{46}$$

The derivative is positive, so the logarithm is strictly increasing. To obtain its multiplication law, fix $a > 0$ and differentiate $\ln(ax) - \ln x$. The derivative vanishes. Evaluating the constant at $x = 1$ gives

$$\ln(ax) = \ln a + \ln x, \qquad \ln 1 = 0, \qquad \ln(1/x) = -\ln x. \tag{47}$$

Since $\ln 2 > 0$, the values $\ln(2^n) = n \ln 2$ are unbounded above. The reciprocal identity supplies unbounded negative values. Continuity and the Intermediate Value Theorem show that the logarithm maps $(0, \infty)$ onto $\mathbb{R}$.

10.2. The exponential as the inverse

Define E to be the inverse of the logarithm. The inverse derivative rule gives $E'(x) = E(x)$ and $E(0) = 1$. The multiplication identity becomes $E(s + t) = E(s)E(t)$. Write $e = E(1)$ and use e^x for $E(x)$. This notation is consistent with integer powers through the addition law.

For $a > 0$, define $a^x = e^{x \ln a}$. The chain rule now establishes the general exponential and positive-base power rules:

$$\frac{\mathrm{d}}{\mathrm{d}x} a^x = a^x \ln a, \qquad \frac{\mathrm{d}}{\mathrm{d}x} x^p = p x^{p-1} \quad (x > 0,\ p \in \mathbb{R}). \tag{48}$$

The real-power restriction avoids assuming that a negative number has every real power. Special integer or rational exponents can have larger real domains, but those domains must be specified separately.

The derivative of $\ln |x|$ is $1/x$ on either side of zero. Consequently, $\int 1/x \, \mathrm{d}x = \ln |x| + C$ on any interval avoiding zero. There can be different constants on the two disconnected intervals. An antiderivative expression does not bridge the singularity.

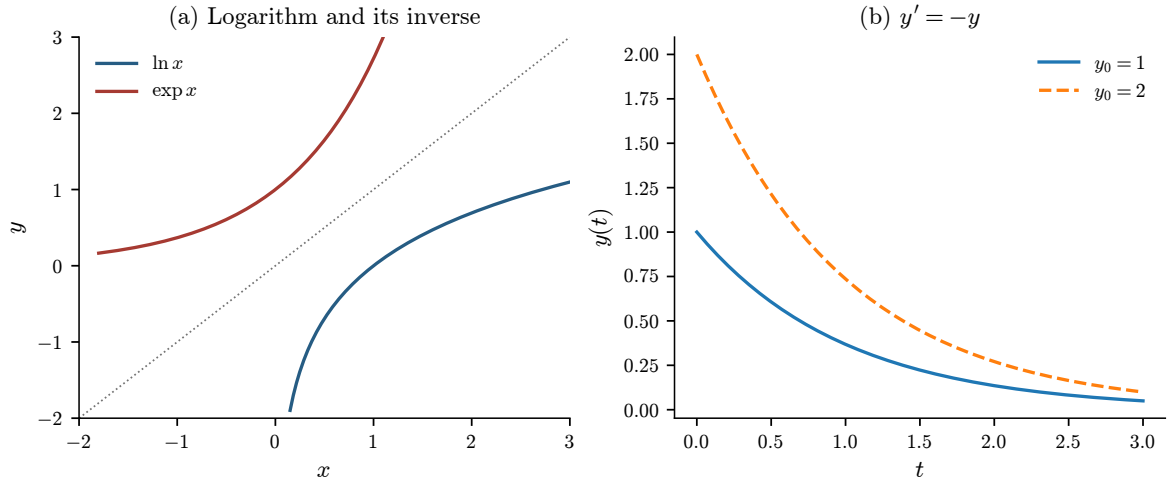


Figure 8. Calculated logarithm and exponential relations. Left: inverse graphs reflected in $y = x$. Right: two exact solutions of $y' = -y$, selected by different initial values. These curves represent stated mathematical models, not fitted observations.

10.3. A rate law and its initial condition

For constant k, the equation $y' = ky$ says that the instantaneous rate is proportional to the current value. The product rule gives $(ye^{-kt})' = 0$. Thus every differentiable solution on an interval containing zero is

$$y(t) = y_0 e^{kt}, \qquad y(0) = y_0. \tag{49}$$

This proof includes $y_0 = 0$ without dividing by y. A negative k gives decay and a positive k gives growth. The parameter has dimensions of inverse time if t is a physical time. A rate law inferred from measurements would need an additional empirical justification; no such inference is being made here.

For a separable equation $y' = g(t)h(y)$, restrict first to a solution range where $h(y) \neq 0$. Choose $H'(y) = 1/h(y)$. The chain rule and endpoint integration give

$$H(y(t)) - H(y(t_0)) = \int_{t_0}^{t} g(s)\,\mathrm{d}s. \tag{50}$$

Zeros of h must be checked separately as possible constant solutions. The algebraic separation does not itself prove existence or uniqueness for an arbitrary rate law. For example, on $t \geq 0$, $y' = 2\sqrt{y}$ with $y(0) = 0$ admits both the zero solution and a solution that remains zero until a chosen time $a \geq 0$, then equals $(t-a)^2$. Both satisfy the equation at the joining time. Additional hypotheses are needed for a general uniqueness theorem.

11. Methods of integration

11.1. Substitution and the transformed interval

The chain rule can be reversed through the Fundamental Theorem. Let f be continuous on a range containing $g([a,b])$ and let g be continuously differentiable. If $F' = f$, then $(F \circ g)' = f(g)g'$. Endpoint evaluation proves

$$\int_a^b f(g(x))g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u. \tag{51}$$

The derivative factor describes how the integration variable changes. The endpoint changes describe the interval traversed by that variable. Both are needed. This oriented identity does not require g to be one-to-one. If g moves forward and backward through the same values, the signed contributions cancel accordingly. Rewriting an integral by an inverse branch is a different procedure and may require splitting the domain.

For instance, with $u = 1 + x^2$,

$$\int_0^1 \frac{2x}{1+x^2}\,\mathrm{d}x = \int_1^2 \frac{1}{u}\,\mathrm{d}u = \ln 2. \tag{52}$$

The relation $\mathrm{d}u = 2x\,\mathrm{d}x$ is shorthand for the derivative factor in the proved substitution formula. It need not be justified by treating differentials as arbitrary independently cancellable real numbers.

11.2. Integration by parts

Integrate the product rule for continuously differentiable u and v:

$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x. \tag{53}$$

This trades one integral for another. A useful choice simplifies one factor when differentiated while leaving an accessible antiderivative for the other. Taking $u = x$ and $v' = e^x$ gives $\int xe^x\,\mathrm{d}x = xe^x - e^x + C$. Taking $u = \ln x$ and $v' = 1$ gives $\int \ln x\,\mathrm{d}x = x\ln x - x + C$ on $x > 0$.

The boundary term is part of a definite calculation. It may contain the entire answer or determine whether an improper limit exists. Removing it because the new integrand looks simpler would change the identity.

11.3. Trigonometric powers and substitution

The choice of technique depends on the structure of the integrand. An odd sine power can retain one sine factor for substitution, using $\sin^2 x = 1 - \cos^2 x$. For example,

$$\int \sin^3 x\,\mathrm{d}x = -\cos x + \frac{\cos^3 x}{3} + C. \tag{54}$$

Even powers can use angle-doubling identities. Since $\sin^2 x = (1 - \cos 2x)/2$,

$$\int \sin^2 x\,\mathrm{d}x = \frac{x}{2} - \frac{\sin 2x}{4} + C. \tag{55}$$

A square root such as $\sqrt{a^2 - x^2}$ suggests $x = a\sin\theta$ for $a > 0$. Restricting θ to $[-\pi/2, \pi/2]$ makes $\cos\theta \geq 0$, so the square root is $a\cos\theta$ rather than an unexamined signed expression. Integration gives

$$\int \sqrt{a^2 - x^2}\,\mathrm{d}x = \frac{1}{2}\left[x\sqrt{a^2 - x^2} + a^2 \arcsin(x/a)\right] + C. \tag{56}$$

The calculation applies on $(-a, a)$, with endpoint values available by continuity for the corresponding definite integral. The restriction is what makes the inverse substitution and square-root sign consistent.

11.4. Rational functions and partial fractions

A rational integrand is first divided if the numerator degree is at least the denominator degree. The proper fraction can then be decomposed according to real linear and irreducible quadratic factors, including all powers of repeated factors. The decomposition is an algebraic identity on the domain where the denominator is nonzero.

For a simple example,

$$\frac{1}{x^2 - 1} = \frac{1}{2}\left(\frac{1}{x - 1} - \frac{1}{x + 1}\right), \qquad \int \frac{\mathrm{d}x}{x^2 - 1} = \frac{1}{2}\ln\left|\frac{x - 1}{x + 1}\right| + C. \tag{57}$$

Each connected interval excluding ± 1 has its own antiderivative constant. An irreducible quadratic instead produces logarithmic or inverse-tangent contributions. Completing the square gives

$$\int \frac{\mathrm{d}x}{x^2+2x+5} = \frac{1}{2}\arctan\left(\frac{x+1}{2}\right) + C. \tag{58}$$

These techniques organize broad families of integrals, but they do not guarantee that every continuous function has an elementary antiderivative. The definite-integral definition and numerical methods remain available when symbolic formulas do not close.

12. Geometric and weighted accumulation

12.1. Area and average value

If continuous functions satisfy $f \geq g$ on $[a,b]$, the area between their graphs is $\int_a^b (f-g)\,\mathrm{d}x$. If their order changes, split at the crossings or integrate $|f-g|$. The signed integral without this correction can cancel distinct positive areas.

For $f(x) = x$ and $g(x) = x^2$ on $[0,1]$, the order is fixed and

$$\mathcal{A} = \int_0^1 (x - x^2)\,\mathrm{d}x = \frac{1}{6}. \tag{59}$$

The average value of an integrable function on a nondegenerate interval is

$$f_{\mathrm{av}} = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x. \tag{60}$$

For continuous f, its minimum and maximum bound the integral, so they also bound this average. The Intermediate Value Theorem supplies a point c with $f(c) = f_{\mathrm{av}}$. This is the mean-value theorem for integrals. The normalization by interval length is essential; without it the integral is a total, with different units.

12.2. Volumes from cross-sections

Suppose a solid has cross-sectional area $A(x)$ perpendicular to the x-axis. Approximating short pieces by cylinders suggests $\sum A(\xi_i)\Delta x_i$. For the regular solids considered here, passage to the integral gives $V = \int_a^b A(x)\,\mathrm{d}x$. The cross-section must represent the actual solid once, without overlapping or omitting pieces.

If $f \geq g \geq 0$ and the region between them rotates about the x-axis, each cross-section is a washer. Rotating about the y-axis instead permits cylindrical shells when $x \geq 0$:

$$V_x = \pi\int_a^b [f(x)^2 - g(x)^2]\,\mathrm{d}x, \qquad V_y = 2\pi\int_a^b x[f(x)-g(x)]\,\mathrm{d}x. \tag{61}$$

The two expressions describe different solids. For the preceding region between x and x^2, they give $V_x = 2\pi/15$ and $V_y = \pi/6$. A radius is a distance from the chosen axis. Rotating a region that crosses the axis requires a fresh cross-section analysis rather than inserting signed radii into a memorized formula.

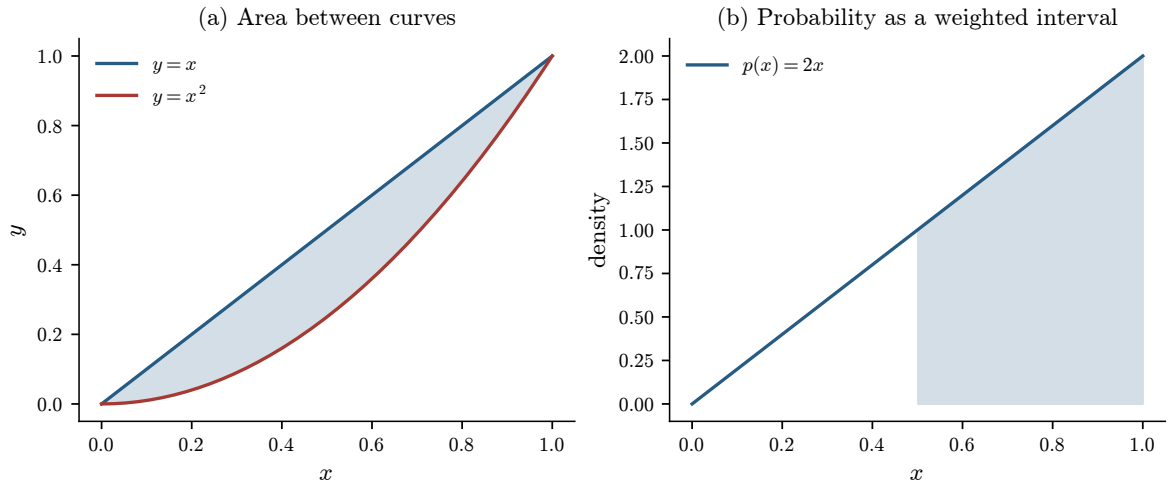


Figure 9. Calculated area and probability examples. Left: the region between $y = x$ and $y = x^2$ on $[0, 1]$, with area $1/6$. Right: the normalized density $p(x) = 2x$ on $[0, 1]$. The shaded interval $x \geq 1/2$ has probability $3/4$. The density is a chosen mathematical example, not an empirical distribution.

12.3. Mass, work and probability

For a rod with nonnegative linear density $\lambda(x)$, its mass is $M = \int_a^b \lambda(x)\,\mathrm{d}x$. If $M > 0$, the centre of mass is the weighted average

$$\bar{x} = \frac{1}{M}\int_a^b x\lambda(x)\,\mathrm{d}x. \tag{62}$$

The same weighted-sum structure appears in probability. A nonnegative density p has total integral one, and its integral over an interval gives the probability assigned to that interval. For $p(x) = 2x$ on $[0, 1]$, the total is one, $P(X \geq 1/2) = 3/4$, and the mean is $\int_0^1 xp(x)\,\mathrm{d}x = 2/3$. The density's value at a point is not itself the probability of that point; an interval width is still needed in the approximating sum.

A variable force supplies another accumulation. Along a line, work is $\int F(x)\,\mathrm{d}x$. For a spring with restoring-force magnitude kx, quasistatic extension from 0 to b requires external work $kb^2/2$. The integral adds the changing force over distance. Multiplying the final force by the full distance would double this value because the force was smaller during the earlier extension.

13. Parametric and polar curves

13.1. A curve need not be a single graph

Represent a plane path by two functions of one parameter, $x = x(t)$ and $y = y(t)$. This is still a one-variable calculation. Suppose the coordinate functions are continuously differentiable. Where $x'(t) \neq 0$, the inverse and chain rules give the first relation below.

The second requires twice differentiable coordinate functions:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y'(t)}{x'(t)}, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{1}{x'(t)}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{y'(t)}{x'(t)}\right). \tag{63}$$

If $x' = 0$ and $y' \neq 0$, the tangent is vertical. If both vanish, neither formula decides the geometry, and the local parameterization needs further analysis. Different parameters can traverse the same curve at different speeds, or trace it more than once.

For a continuously differentiable regular curve, polygonal length sums converge to

$$L = \int_{t_0}^{t_1} \sqrt{x'(t)^2 + y'(t)^2}\,\mathrm{d}t. \tag{64}$$

The integrand follows by applying the local linear model to both coordinate increments and using the Euclidean distance formula. Standard rectifiable-curve theory justifies the limiting length identity for the stated regularity; its full partition proof is not repeated here. An isolated zero speed at an endpoint can be handled by a limiting or piecewise argument.

13.2. A cycloid and a surface of revolution

One unit-radius cycloid arch is $x = t - \sin t$, $y = 1 - \cos t$ for $0 \le t \le 2\pi$. Its speed is

$$\sqrt{(1-\cos t)^2 + \sin^2 t} = 2\sin(t/2), \qquad L = \int_0^{2\pi} 2\sin(t/2)\,\mathrm{d}t = 8. \tag{65}$$

The sine is nonnegative over this interval; extending the speed formula to other intervals would require its absolute value. The parameter speed vanishes at the cusped endpoints, while the interior is regular.

For a regular curve above the x-axis, rotation around that axis gives surface area

$$S = 2\pi \int_{t_0}^{t_1} y(t)\sqrt{x'(t)^2 + y'(t)^2}\,\mathrm{d}t. \tag{66}$$

The approximation uses short conical frusta, whose areas tend to circumference times arc length. The formula assumes the surface is counted once and $y \ge 0$ is the radius. It concerns a surface area, whereas the preceding washer integral concerns an enclosed volume.

13.3. Polar area and its traversal

For $x = r(\theta)\cos\theta$ and $y = r(\theta)\sin\theta$, a short angular sector has area approximately $r^2\Delta\theta/2$. When a continuous nonnegative radial graph traces the intended region once over $[\alpha, \beta]$, the area is

$$\mathcal{A} = \frac{1}{2}\int_{\alpha}^{\beta} r(\theta)^2\,\mathrm{d}\theta. \tag{67}$$

The factor $1/2$ comes from sector geometry in radians. A polar curve may loop, cross the origin or retrace a region, so the angular interval is part of the geometric specification. The integral can otherwise count area more than once.

For the cardioid $r = 1 + \cos\theta$ on $[0, 2\pi]$, the radius is nonnegative and the enclosed region is traced once. Using the integral of $\cos^2\theta$ gives

$$\mathcal{A} = \frac{1}{2}\int_0^{2\pi} (1 + \cos\theta)^2 \,\mathrm{d}\theta = \frac{3\pi}{2}. \tag{68}$$

For a continuously differentiable radius, differentiating the Cartesian coordinates gives the corresponding polar length formula

$$L = \int_\alpha^\beta \sqrt{r(\theta)^2 + r'(\theta)^2}\,\mathrm{d}\theta. \tag{69}$$

It follows because the cross terms cancel in $x'^2 + y'^2$. Both polar area and length therefore remain one-variable integrals, although they encode different local geometric quantities.

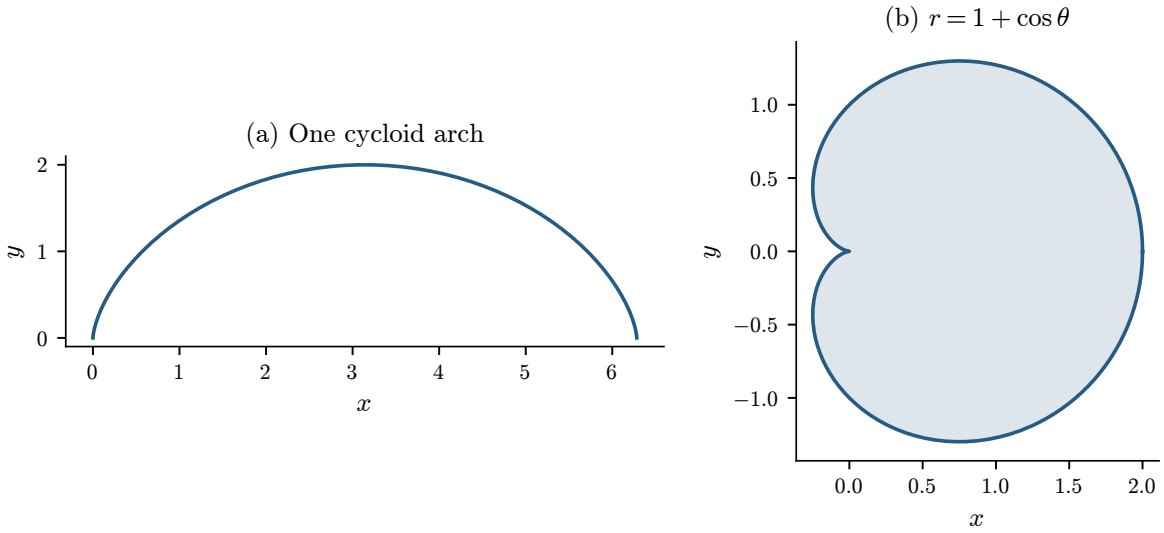


Figure 10. Calculated non-Cartesian descriptions. Left: one unit-radius cycloid arch, of length 8. Right: the cardioid $r = 1 + \cos\theta$, whose enclosed area is $3\pi/2$. The axes have equal scale within each panel.

14. Improper limits and singular integrands

14.1. An infinite endpoint is a new limit

The ordinary Riemann definition uses a bounded function on a finite interval. An infinite interval requires an additional limit:

$$\int_a^\infty f(x)\,\mathrm{d}x = \lim_{B\to\infty}\int_a^B f(x)\,\mathrm{d}x, \tag{70}$$

provided the finite integrals exist and this limit is finite. An unbounded integrand at a finite endpoint is treated similarly, for instance $\int_0^1 f = \lim_{\epsilon\downarrow 0}\int_\epsilon^1 f$. Calling the expression an integral does not establish convergence.

For the power family, endpoint evaluation gives two distinct thresholds:

$$\int_1^\infty x^{-p}\,\mathrm{d}x = \frac{1}{p-1} \quad (p>1), \qquad \int_0^1 x^{-p}\,\mathrm{d}x = \frac{1}{1-p} \quad (p<1). \tag{71}$$

The first diverges for $p \le 1$ and the second for $p \ge 1$. At $p = 1$, the logarithm provides the divergent endpoint behaviour. A function can decay to zero and still accumulate an infinite total, as $1/x$ at infinity demonstrates. An unbounded function can have finite improper area, as $x^{-1/2}$ near zero demonstrates.

14.2. Comparison and a singularity inside an interval

If $0 \le f \le g$ sufficiently far out and $\int g$ converges, the increasing truncated integrals of f are bounded above. Completeness supplies their finite limit. Conversely, a divergent nonnegative lower comparison proves divergence of a larger integrand. A finite initial interval does not affect convergence of the tail.

Remark 14.1 (Principal value is a different limit). An interior singularity at c requires two independently convergent one-sided integrals. For $1/x$ on $[-1,1]$, the left side tends to $-\infty$ and the right side to $+\infty$. The improper integral does not exist. Symmetric deletion instead gives

$$\lim_{\epsilon\downarrow 0}\left(\int_{-1}^{-\epsilon}\frac{\mathrm{d}x}{x} + \int_\epsilon^1 \frac{\mathrm{d}x}{x}\right) = 0. \tag{72}$$

That is a Cauchy principal value. Its coordinated cancellation is a different limiting operation from two independent improper integrals. Endpoint antiderivative evaluation across zero would erase precisely this distinction.

14.3. L'Hospital's rule and its hypotheses

For a one-sided $0/0$ limit, suppose f and g are continuous at the limiting endpoint with common value zero, differentiable on a punctured interval, and $g' \neq 0$ there. If $f'/g' \to L$, the Cauchy Mean Value Theorem gives

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)} \to L, \tag{73}$$

where c_x lies between the endpoint and x. The Cauchy theorem itself follows by applying Rolle's theorem to a linear combination of f and g chosen to have equal endpoint values. The nonzero derivative condition makes g strictly monotone locally and permits the division.

There is also an ∞/∞ version under the corresponding differentiability, denominator and derivative-ratio hypotheses. That extension is used as an established limit theorem here. The rule does not mean that every quotient can be replaced by

the quotient of its derivatives. For example, $\lim_{x\to 0}(1+x)/1 = 1$ is an ordinary limit, whereas differentiating the denominator would create an undefined quotient.

With the exponential already constructed, two valid applications give

$$\lim_{x\to 0}\frac{e^x-1}{x}=1, \qquad \lim_{x\to\infty}\frac{\ln x}{x}=0. \tag{74}$$

The first relation also follows directly from the derivative of the exponential at zero. Using the rule to establish $\sin x/x \to 1$ would be circular if the sine derivative had itself been derived from that limit. The order of the arguments therefore matters even when the final calculation is short.

15. Infinite series and convergence

15.1. A sum means a limit of partial sums

For a sequence (a_n), define $S_N = \sum_{n=1}^{N} a_n$. The series $\sum_{n=1}^{\infty} a_n$ converges to S when $S_N \to S$. A necessary condition is $a_n \to 0$, because $a_n = S_n - S_{n-1}$. It is not sufficient. Grouping the harmonic series into blocks gives

$$\sum_{n=2^{k-1}+1}^{2^k}\frac{1}{n} \geq 2^{k-1}\frac{1}{2^k}=\frac{1}{2}. \tag{75}$$

Each new block contributes at least one-half, so the partial sums are unbounded. Every individual term tends to zero, but the accumulated tail does not become small. The geometric series behaves differently because its tail has a vanishing explicit bound.

For a series with nonnegative terms, the partial sums are increasing. The series converges precisely when those sums are bounded above, by completeness. This gives a framework in which comparison tests are statements about bounds on finite sums, followed by a justified limit.

15.2. Comparison, ratios and roots

If $0 \leq a_n \leq b_n$ eventually and $\sum b_n$ converges, the partial sums of a_n are bounded and therefore converge. A divergent smaller nonnegative series proves divergence of a larger one. If $a_n/b_n \to c$ with $0 < c < \infty$, eventual two-sided bounds give the limit-comparison test.

For decreasing positive f, rectangles compare $\sum f(n)$ with $\int f(x)\,\mathrm{d}x$. The integral test applied to $f(x) = x^{-p}$ gives convergence of $\sum n^{-p}$ exactly when $p > 1$. For $p \leq 0$, even the term condition fails. When $p = 2$, the omitted tail obeys

$$\frac{1}{N+1} \leq \sum_{n=N+1}^{\infty}\frac{1}{n^2} \leq \frac{1}{N}. \tag{76}$$

This estimates an error without requiring a closed form for the full sum.

If the terms are eventually nonzero and $|a_{n+1}/a_n| \to q < 1$, choose $q < r < 1$. Every sufficiently late ratio is at most r, so the absolute tail is bounded by a geometric

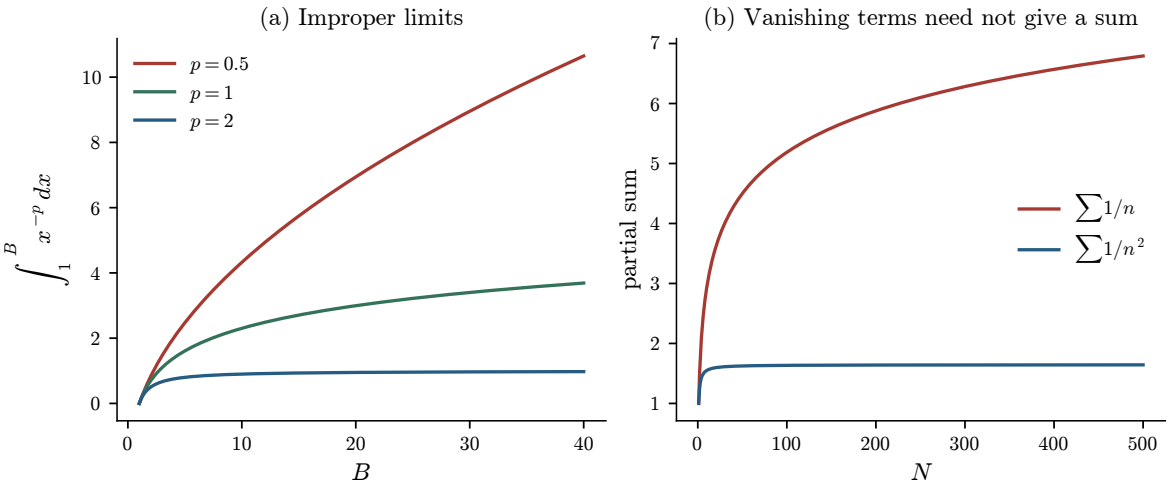


Figure 11. Calculated partial accumulation. Left: integrals of x^{-p} from 1 to a finite endpoint B. Right: partial sums of $1/n$ and $1/n^2$. Every displayed curve is finite. The comparison and block arguments, rather than the finite plots, determine convergence or divergence.

series. The series converges absolutely. If the ratio limit exceeds one, the terms fail to approach zero. A ratio limit of one gives no decision: both $1/n$ and $1/n^2$ have that limit.

The root test compares $|a_n|^{1/n}$ with a geometric ratio. If its limit is less than one there is absolute convergence; if it is greater than one the terms do not tend to zero. The test remains valid with an upper limit, or lim sup, replacing the ordinary limit. Here $\limsup b_n = \lim_{N\to\infty} \sup_{n\ge N} b_n$, allowing extended values. The equality case one is again inconclusive.

15.3. Absolute convergence and cancellation

If $\sum |a_n|$ converges, then any tail of $\sum a_n$ is bounded in magnitude by the corresponding absolute tail. The partial sums are Cauchy and converge in $\mathbb{R}$. Such a series is absolutely convergent. A convergent series whose absolute series diverges is conditionally convergent.

For an alternating series $b_1 - b_2 + b_3 - \cdots$ with $b_n \ge 0$, b_n nonincreasing and $b_n \to 0$, the even partial sums rise and the odd partial sums fall. They bound one another, and their gap tends to zero. Completeness gives a common limit. The remainder after N terms has magnitude at most b_{N+1}.

The alternating harmonic series is therefore convergent although the harmonic series of its absolute values diverges. Its value is $\ln 2$. One way to verify the value is to integrate the finite identity

$$\frac{1}{1+t} = \sum_{k=0}^{N-1} (-t)^k + \frac{(-t)^N}{1+t}, \qquad 0 \le t \le 1. \tag{77}$$

The remainder integral has magnitude at most $\int_0^1 t^N \,\mathrm{d}t = 1/(N+1)$, which tends to zero. This argument identifies the limit without assuming that an infinite series may automatically be integrated term by term at an endpoint.

15.4. Pointwise and uniform convergence

A sequence of functions can converge at every fixed input while lacking a common error bound across an interval. For $f_n(x) = x^n$ on $[0,1]$, the pointwise limit is zero for $x < 1$ and one at $x = 1$. Each f_n is continuous, while the limit is not. The convergence is not uniform.

Uniform convergence to f means that the largest error tends to zero:

$$\sup_{x \in E} |f_n(x) - f(x)| \to 0. \tag{78}$$

Theorem 15.1 (Continuity under uniform convergence)**.** *The uniform limit of continuous real-valued functions on a set $E \subseteq \mathbb{R}$ is continuous on E.*

Proof. To prove continuity at $a \in E$, choose one sufficiently large N so that both $|f - f_N|$ errors are small, then use continuity of f_N to control the middle term in

$$|f(x) - f(a)| \le |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)|. \tag{79}$$

□

Theorem 15.2 (Integration under uniform convergence)**.** *If Riemann-integrable functions f_n on $[a,b]$ converge uniformly to f, then f is Riemann integrable and*

$$\left| \int_a^b f_n \,\mathrm{d}x - \int_a^b f \,\mathrm{d}x \right| \le (b-a) \sup_{[a,b]} |f_n - f| \to 0. \tag{80}$$

Proof. Uniform closeness to a bounded f_N makes f bounded. The difference between its upper and lower sums is at most the corresponding gap for f_N plus $2(b-a)\sup|f - f_N|$. First choose N to make the uniform error small, then a partition to make the gap for f_N small. The Darboux criterion gives integrability. Integrating the uniform absolute-error bound gives the displayed inequality and hence convergence of the integrals. □

Remark 15.3 (Uniform convergence does not justify differentiating). Differentiation needs more. The functions $\sin(nx)/n$ tend uniformly to zero, but their derivatives $\cos(nx)$ do not tend to the derivative zero; at $x = 0$ they are always one. A sufficient theorem assumes uniformly convergent continuous derivatives together with convergence of the function values at one point. The Fundamental Theorem then constructs the limiting function by integrating the limiting derivative. It is the derivative control, rather than the graph's small height alone, that supports the exchange.

16. Taylor approximation and power series

16.1. Matching derivatives at one point

A linear approximation matches the value and first derivative. A degree-n Taylor polynomial matches the derivatives through order n at a centre a:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k. \tag{81}$$

The factorial appears because differentiating $(x-a)^k$ exactly k times produces $k!$, while lower-order derivatives of that term vanish at a. These coefficients are forced by the matching conditions. They are not chosen by a numerical fit over an interval.

Matching derivatives does not yet bound the difference away from a. Write $f(x) = T_n(x) + R_n(x)$ and retain the remainder until its behaviour has been established.

16.2. Taylor's theorem with a remainder

Theorem 16.1 (Taylor's theorem with Lagrange remainder)**.** *Let $n \geq 0$ be an integer and let f have continuous derivatives through order $n+1$ on an interval containing distinct points a and x. For the degree-n Taylor polynomial at a, there is a point ξ strictly between a and x such that*

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}. \tag{82}$$

Proof. Choose C so that $\psi(t) = f(t) - T_n(t) - C(t-a)^{n+1}$ vanishes at $t = x$. At a, ψ and its first n derivatives vanish. Repeated application of Rolle's theorem yields $\psi^{(n+1)}(\xi) = 0$. Since the Taylor polynomial has zero $(n+1)$st derivative, this identifies $C = f^{(n+1)}(\xi)/(n+1)!$ and proves the formula. □

If $|f^{(n+1)}| \leq M$ between the endpoints, the computable consequence is

$$|R_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}. \tag{83}$$

The point ξ need not be known. A valid bound on the derivative over the whole intervening interval is enough.

Example 16.2 (A certified square-root estimate)**.** For $f(x) = \sqrt{x}$ near 4, $f''(x) = -1/(4x^{3/2})$. On $[4, 4.04]$ its magnitude is at most $1/32$. Thus the linear estimate 2.01 for $\sqrt{4.04}$ has error at most $(0.04)^2/64 = 0.000025$. The concavity shows that the estimate is above the exact value.

16.3. When the infinite polynomial represents the function

The Taylor series represents f at x only if $R_n(x) \to 0$ as $n \to \infty$. For the exponential on a fixed finite interval, every derivative equals the exponential and is bounded by a constant independent of n. The factorial bound makes the remainder tend to zero. For sine and cosine, all derivative magnitudes are at most one. Therefore

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \qquad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}. \tag{84}$$

These identities hold for every real x. The factorial denominator dominates any fixed power, which can be proved by a ratio comparison. For $\sin x$, using its degree-seven polynomial on $|x| \le 1$ and treating the vanishing eighth coefficient explicitly gives $|R| \le 1/9!$. A low-degree approximation thus comes with an interval and an error estimate.

The geometric identity yields $1/(1-x) = \sum_{k=0}^{\infty} x^k$ for $|x| < 1$. Integrating on any smaller closed interval, where the convergence is uniform, gives

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{k} \quad (|x| < 1). \tag{85}$$

At $x = 1$ the series converges to $\ln 2$ by the separate remainder argument already given. At $x = -1$ it diverges. The radius alone does not decide either endpoint.

16.4. The radius of convergence

For a power series $\sum c_n (x-a)^n$, the root test gives radius

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}, \tag{86}$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$. There is absolute convergence for $|x-a| < R$ and divergence for $|x-a| > R$. Each endpoint at distance R must be tested independently.

On any closed interval $|x-a| \le r < R$, absolute geometric majorants give uniform convergence. The differentiated series has the same radius, because the additional factor n does not change the limiting nth-root growth rate. Uniform convergence of that derivative series on smaller intervals justifies termwise differentiation. Termwise integration follows from the earlier integral bound. These operations are controlled inside the radius, without a blanket permission to exchange arbitrary infinite operations.

16.5. Smoothness is not analyticity

Example 16.3 (A smooth function not represented by its Taylor series)**.** Define $f(0) = 0$ and $f(x) = e^{-1/x^2}$ for $x \neq 0$. Every derivative away from zero is the exponential times a polynomial in $1/x$. For every fixed nonnegative integer m,

$$\lim_{x\to 0} |x|^{-m} e^{-1/x^2} = 0. \tag{87}$$

To verify the limit, put $u = 1/x^2$ and compare $u^{m/2}$ with e^u; an integer power above $m/2$ is enough for repeated L'Hospital differentiation or a positive exponential-series bound. The exponential wins over every such fixed power.

Induction, using the derivative difference quotient at zero at each stage, now gives $f^{(k)}(0) = 0$ for every k. Its Taylor series at zero is identically zero, although $f(x) > 0$ whenever $x \neq 0$. The function is smooth but not analytic at zero. Its complete derivative data at that point do not determine its values in a neighbourhood through a Taylor series. The missing information resides in the remainder.

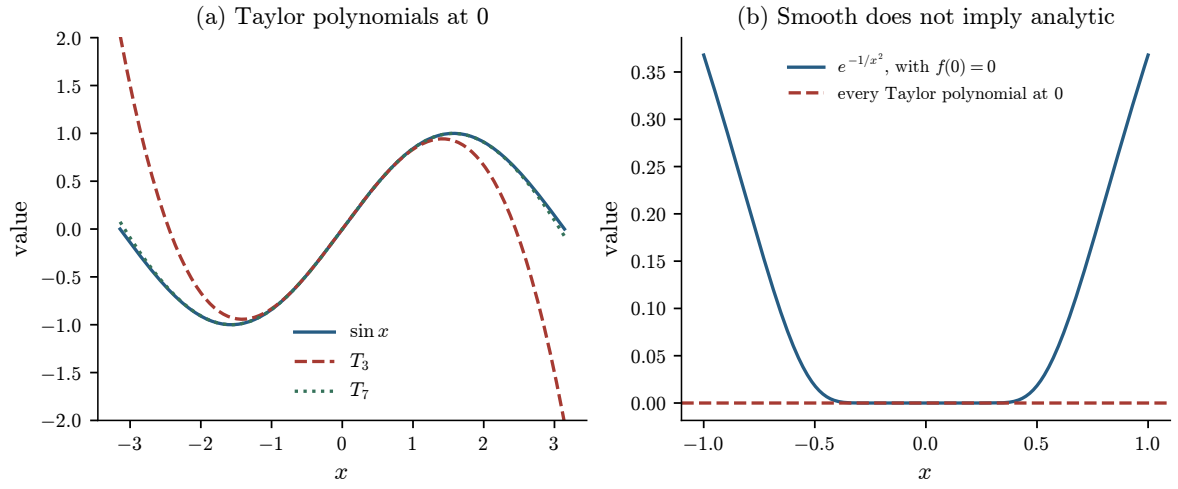


Figure 12. Calculated Taylor examples. Left: sine and its degree-three and degree-seven Taylor polynomials at zero. Right: the smooth function e^{-1/x^2}, extended by zero at the origin, and its identically zero Taylor polynomials there. Agreement of every derivative at one point does not by itself imply a local series representation.

17. Numerical approximation and its error

17.1. Finite differences in finite arithmetic

The derivative is a limit, while a computer evaluates a quotient at a finite increment. For a smooth function, Taylor expansion gives the forward and central approximations

$$D_h^+ f(a) = \frac{f(a+h) - f(a)}{h}, \qquad D_h^0 f(a) = \frac{f(a+h) - f(a-h)}{2h}. \tag{88}$$

The forward truncation error is $O(h)$. Symmetry cancels its first-order contribution in the central formula, giving $O(h^2)$ when the third derivative is bounded nearby. Specifically, bounds $|f''| \leq M_2$ and $|f'''| \leq M_3$ give errors at most $M_2|h|/2$ and $M_3 h^2/6$, respectively, under the corresponding Taylor hypotheses.

Floating-point arithmetic adds a different error. Subtracting nearby function values can remove leading digits; dividing the remaining rounding error by h amplifies it. In the usual rounding model this contribution can scale like $\epsilon_{\text{mach}}/|h|$ times a function-value

scale. The model explains why reducing h indefinitely need not improve the computed quotient. It is not a universal exact error formula for every function or implementation.

The accompanying calculation differentiates $\sin x$ at $x = 1$, using 161 logarithmically spaced increments from 10^{-1} to 10^{-17} in NumPy binary64 arithmetic. Errors are compared with the computed value $\cos 1$. Both the approximations and their reference value have finite arithmetic precision. The graph shows the expected initial truncation trend and eventual loss of accuracy. It is a numerical illustration of competing errors, not an experimental measurement or a proof about every computer.

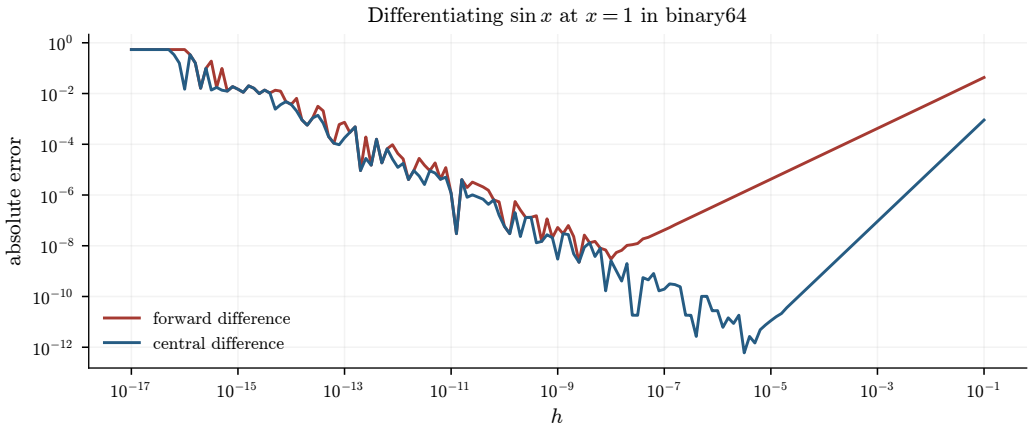


Figure 13. Computed absolute errors for forward and central differences of $\sin x$ at 1. The horizontal variable is a finite increment, not an infinitesimal. The reference is the binary64 evaluation of $\cos 1$. At very small increments, rounding and subtraction dominate the truncation improvement.

17.2. Quadrature rules and regularity

For n equal subintervals of width $h = (b - a)/n$, the midpoint and trapezoid approximations are

$$M_n = h \sum_{i=1}^{n} f\left(a + (i - \frac{1}{2})h\right), \qquad T_n = h\left[\frac{f(a) + f(b)}{2} + \sum_{i=1}^{n-1} f(a + ih)\right]. \tag{89}$$

They approximate each short piece by a constant or a line. Taylor's theorem about each midpoint bounds the local midpoint error by $M_2 h^3/24$; summing the errors gives the global estimate. Linear-interpolation error gives the corresponding trapezoid estimate:

$$|I - M_n| \le \frac{M_2(b-a)^3}{24n^2}, \qquad |I - T_n| \le \frac{M_2(b-a)^3}{12n^2}, \qquad |f''| \le M_2. \tag{90}$$

Composite Simpson quadrature integrates a quadratic interpolant across each pair of subintervals. With even n, its endpoint, odd-index and even-index weights are

$$S_n = \frac{h}{3}\left[f(a) + f(b) + 4\sum_{j=1}^{n/2} f(a + (2j-1)h) + 2\sum_{j=1}^{n/2-1} f(a + 2jh)\right]. \tag{91}$$

Here the displayed sums simply select odd and even interior nodes. The classical interpolation error theorem gives

$$|I - S_n| \le \frac{M_4(b-a)^5}{180n^4}, \qquad |f^{(4)}| \le M_4, \tag{92}$$

for a function with continuous fourth derivative. This bound is used as an established quadrature theorem; its full interpolation remainder proof is not reproduced. The regularity assumption and even-subdivision condition are retained. A singular integrand cannot inherit fourth-order behaviour merely because Simpson weights were applied to it.

17.3. A reproducible convergence calculation

Compute $I = \int_0^1 (1+x)^{-1}\,\mathrm{d}x = \ln 2$ with $n = 4, 8, 16, 32, 64$. The same smooth integrand and interval are used for all rules. The following table reports absolute errors against the binary64 logarithm evaluation, rounded from the stored computations.

Table 1. Computed quadrature errors for the stated logarithm integral.

n	Midpoint error	Trapezoid error	Simpson error
4	1.93×10^{-3}	3.88×10^{-3}	1.07×10^{-4}
8	4.87×10^{-4}	9.75×10^{-4}	7.35×10^{-6}
16	1.22×10^{-4}	2.44×10^{-4}	4.72×10^{-7}
32	3.05×10^{-5}	6.10×10^{-5}	2.97×10^{-8}
64	7.63×10^{-6}	1.53×10^{-5}	1.86×10^{-9}

The midpoint and trapezoid error ratios approach four when n doubles. The Simpson ratios approach sixteen. The first Simpson ratio is not exactly sixteen; the statement concerns the developing asymptotic trend over these actual runs. Agreement with the expected orders checks this implementation on this smooth example. It does not establish the same convergence for singular, discontinuous or arbitrarily oscillatory functions.

The Newton square-root calculation is also recorded using Python decimal arithmetic with precision 70. From $x_0 = 1$, the absolute errors after four and five updates are approximately 1.59×10^{-12} and 8.99×10^{-25}. These values are computed differences from the same-precision square-root reference. Their interpretation is consistent with the exact squared-error relation derived earlier. No uncertainty interval or physical observation is being assigned to them.

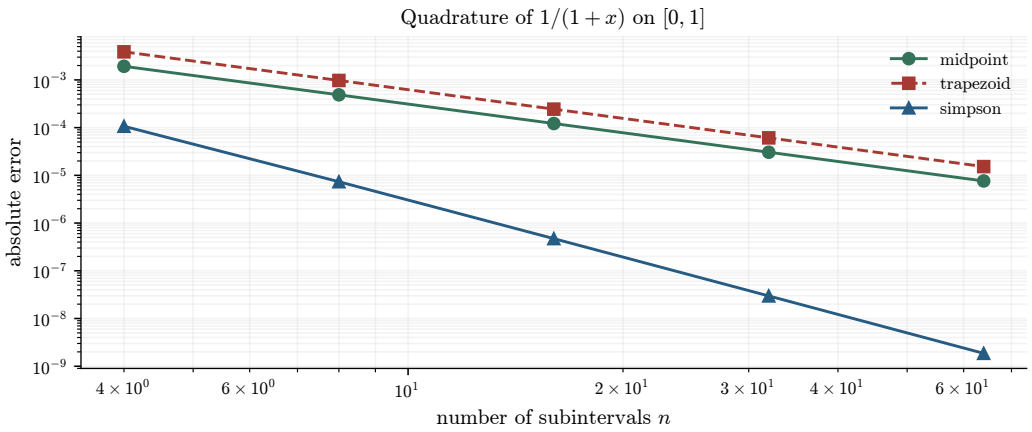


Figure 14. Computed quadrature errors for $(1+x)^{-1}$ on $[0,1]$. The samples and their corresponding CSV data are generated by the supplied Python code. The log–log trends reflect this integrand and the displayed subdivision counts.

18. Conclusions and scope

Calculus turns approximation into an exact statement by controlling what remains in a limit. Completeness supports the existence of suitable limiting values. Differentiability identifies a unique local linear coefficient, while the Mean Value Theorem connects that coefficient to finite changes under interval hypotheses. Riemann integration requires the total sampling error to disappear across all sufficiently fine partitions. The Fundamental Theorem links the derivative of accumulation to its integrand and makes endpoint evaluation possible.

The later methods inherit these conditions. Substitution follows the chain rule and must transform endpoints and derivative factors consistently. Geometric integrals depend on the region, axis and traversal being counted. Improper integrals introduce new endpoint limits, and infinite series require convergence of partial sums. A Taylor polynomial becomes a representation only after the remainder is shown to vanish. Smoothness alone does not supply that conclusion.

The computations illustrate the difference between a mathematical limit and a finite implementation. Small increments compete with rounding error. Quadrature orders appear under suitable smoothness. Newton iteration is locally accurate near a simple root while its global basins can be intricate. The recorded grids, arithmetic, stopping criteria and data make these illustrations reproducible without giving them the status of proofs.

The exposition assumes the ordered complete real field and explicitly states the compactness, quadrature and curve-length results it uses. It does not construct the real numbers or develop Lebesgue integration, global complex dynamics or multivariable calculus. Throughout, a formula remains usable only when its approximation and domain have been accounted for.

Appendix A. Advanced extension: Newton basins in the complex plane

This appendix goes beyond real single-variable calculus. It connects the local real iteration in Section 7 with a finite illustration of complex polynomial dynamics. None of its results is needed for the main development.

A.1. The algebraic extension

The real Newton update can select different roots from different starting values. To display that dependence for a cubic, extend the polynomial arithmetic to $z = x + iy$, where $i^2 = -1$. The pair (x, y) gives the horizontal and vertical coordinates in the complex plane, and $|z| = \sqrt{x^2 + y^2}$. Multiplication and division use ordinary algebra together with $i^2 = -1$.

For $p(z) = z^3 - 1$, the same polynomial differentiation rule gives $p'(z) = 3z^2$. Its Newton map is

$$N(z) = z - \frac{z^3 - 1}{3z^2} = \frac{1}{3}\left(2z + \frac{1}{z^2}\right), \qquad z \neq 0. \tag{A.1}$$

Factoring $(z - 1)(z^2 + z + 1)$ gives three roots:

$$\alpha_0 = 1, \qquad \alpha_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad \alpha_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i. \tag{A.2}$$

For each root, its basin of attraction is the set of starting points whose exact iterates are defined and converge to that root. The basin is defined through an infinite limiting behaviour. A finite image can only sample and approximate its membership.

A.2. Why a simple root attracts nearby iterates

Fix a root α and write $z = \alpha + e$. Because $\alpha^3 = 1$, a direct algebraic calculation gives

$$N(z) - \alpha = \frac{(z - \alpha)^2(2z + \alpha)}{3z^2}. \tag{A.3}$$

Near α, the denominator stays away from zero and the remaining factor stays bounded. The new error is therefore bounded by a constant times $|e|^2$. This is the complex polynomial counterpart of the local real error estimate. It uses no claim that a general real-variable proof automatically establishes every result of complex analysis.

The local estimate does not classify every remote starting point. The pole at zero interrupts the update, and points near a pole can be carried far away. Iterates may eventually return to a root's neighbourhood or remain unresolved within a finite computation. Such possibilities make the global geometry richer than three simple nearest-root regions.

A.3. What the sampled image records

Figure A1 uses a 2000×1250 grid, including endpoints, over $-1.8 \leq \operatorname{Re} z \leq 1.8$ and $-1.125 \leq \operatorname{Im} z \leq 1.125$. Each starting point is iterated in NumPy complex128 arithmetic for at most 60 updates. The calculation checks the initial value and the result after each update. It accepts convergence when $|z^3 - 1| \leq 10^{-10}$ and labels the point by the nearest exact root.

The update is guarded when $|3z^2| \leq 10^{-14}$. Unresolved or guarded points are black. Blue denotes the root 1, orange the root with positive imaginary part, and green its conjugate. Brightness decreases with the number of updates before acceptance. The even grid sizes do not sample zero itself; its undefined update remains an exact algebraic fact rather than a pixel classification in this run.

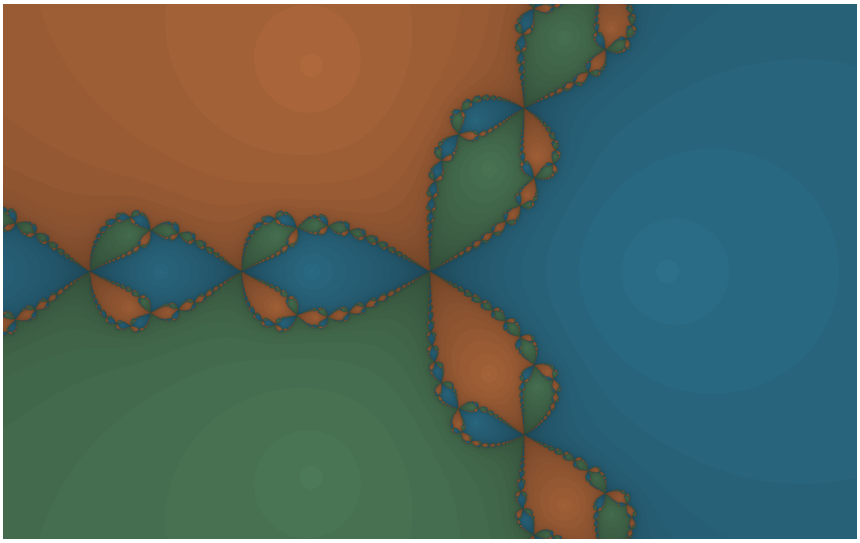


Figure A1. Computed Newton-basin image for $z^3 - 1$, using the window, grid, stopping tolerance and iteration cap stated in the text. Colour identifies the accepted root and brightness encodes update count. The raster retains all 2000×1250 computed pixels. This is a mathematical numerical illustration, not a photograph, an experimental dataset or a proof of the boundary's structure.

In this run 2,499,998 starting pixels meet the residual criterion and two remain unresolved at the cap. No sampled point triggers the derivative guard. These are counts for this particular grid and arithmetic. They do not measure the areas of exact basins, certify every accepted orbit in exact arithmetic or prove that either unresolved orbit fails to converge.

The finely interwoven boundary is commonly called a Newton fractal. Its global dynamical theory lies outside this exposition. The computation separates local convergence from global behaviour. An update that squares a small error near a root can make the eventual root difficult to infer remotely. Greater resolution reveals further sampled detail, but no finite image proves structure at every scale.

Bibliography and source record

The principal course is MIT 18.01SC [1]. Gross [2] and the visual lessons [3] inform the treatment of limits and accumulation. The real-analysis material [4] supports the theorem hypotheses. Numerical methods and the optional appendix also consult [5–7]. This independently written exposition presents established mathematics. Artwork credits appear in [9–11].

The source index retains the original URLs and links from 278 MIT clip pages to 36 distinct full lectures. It also records 38 Gross, 25 Real Analysis and 13 3Blue1Brown transcripts, with matching captions and six supplementary note files. Duplicate transcripts were consolidated, and original transcription errors were preserved. Computations are reproducible from the supplied code. Sources were accessed 31 October 2026.

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